Project: Worked examples
Title: Motor and actuator sizing
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1. Scope and method

Calculates the torque, speed and power a motor needs to move a load through a mechanism, and checks a candidate motor against them. A continuously running rotary load is sized from its steady torque, speed and run-up time.

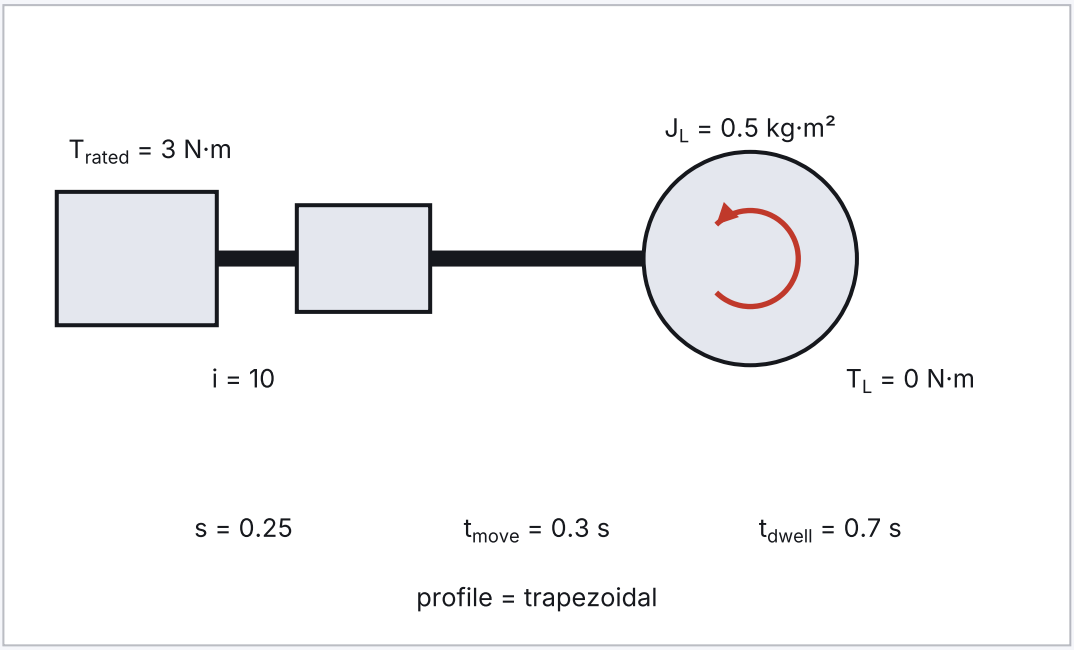


Figure 1 — A motor through a gearbox to a rotary load of inertia J_L and torque T_L. Not to scale; the labels are the inputs as entered.

WHAT IT CALCULATES	RELATION
Linear mechanism	$k = p/(2\pi\, i)$ (screw), $D/(2\, i)$ (belt), $D_p/(2\, i)$ (rack), in metres per motor radian; $J_{refl} = m\, k^2/\eta$; $T_{load} = F\, k/\eta$, with $F = m\, g\, (\sin\theta + \mu \cos\theta) + F_p$
Rotary mechanism through a gearbox	$J_{refl} = J_L/(i^2\, \eta)$; $T_{load} = T_L/(i\, \eta)$
Move profile	trapezoidal $v_{pk} = s/(t_{move} - t_{acc})$, $a = v_{pk}/t_{acc}$; S-curve (sine acceleration, same peak speed) $a_{pk} = (\pi/2)\, a_{trap}$
Peak torque	$T_{acc} = J_{total}\, \alpha$; $T_{pk} = T_{acc} + T_{load}$; $T_{dec} = -T_{acc} + T_{load}$; the check is $SF \cdot \max(\lvert T_{pk}\rvert, \lvert T_{dec}\rvert) \le T_{peak}$; braking power $P_{brake} = \max(0, -T_{dec} \cdot \omega_{pk})$
Thermal duty	$T_{rms} = \sqrt{(\Sigma\, T_i^2\, t_i / t_{cycle})}$ over the whole cycle, the dwell at $T_{hold} = m\, g \sin\theta \cdot k$ for a linear mechanism not held by a brake; $P_{rms} = \omega_{pk}\, \sqrt{\{[t_{acc} \cdot (2/3)(T_{load}^2 + T_{acc}^2) + T_{load}^2\, t_{const}]/t_{cycle}\}}$, and for the S-curve the acceleration term is $t_{acc} \cdot [(3/4)\, T_{load}^2 + (5/16)\, T_{acc}^2]$

WHAT IT CALCULATES	RELATION
Output side and design torque	$T_{alt} = (T_{pk} - T_{dec})/2$ (0 in continuous duty); $T_{out} = T \cdot i \cdot \eta$ and $n_{out} = n_{pk}/i$ for a rotary mechanism; $F_{ax,pk} = F_{load} + m\, a_{pk}$ for a ball screw; T_{design} on the basis T_{basis} (load peak $\max(\lvert T_{pk}\rvert, \lvert T_{dec}\rvert)$, T_{peak}, $SF_c \cdot T_{rated}$ or T_{lim})
Continuous rotary duty	$\omega = 2\pi\, n_L\, i/60$, $n_{pk} = n_L\, i$; $T_{load} = T_L/(i\,\eta)$, $T_{rms} = T_{load}$; $T_{pk} = J_{total}\, \omega/t_{start} + T_{load}$; $P_{cont} = T_{load}\, \omega$, $P_{rms} = P_{cont}$
Rated power of the candidate motor	$P_{rated} = T_{rated} \cdot 2\pi\, n_{rated}/60$ (without n_{rated} there is no P_{rated}); the check is $P_{cont} \le P_{rated}$, $u_{power} = P_{cont}/P_{rated}$

Basis: Load torque, inertia referred to the motor, duty cycles[1](Ch. on motor and drive selection, 5th ed. (edition to be checked)).

2. Inputs

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
mechanism	Mechanism	gearbox rotary		assumption: Sizing convention.	any
$duty_{mode}$	Duty (point-to-point move cycle, or continuous rotary running with a run-up)	point-to-point		spec: Cycle specification CS-7: 90° index then dwell	any
profile	Motion profile	trapezoidal		assumption: Sizing convention.	any
T_{basis}	Design torque basis for downstream couplings (T_{design}): the load's peak (the larger of the accelerating and the braking torque, in magnitude), the motor's typed peak torque, a service factor on the rated torque, or a protective limit such as a torque limiter	Load peak (T_pk)		assumption: Design torque for the coupling taken at the load peak; template default.	any
brake	A brake or a self-locking drive holds a linear axis during the dwell (no holding torque); unticked, a vertical or inclined axis is held by the motor	no		assumption: Linear axes only; the rotary table does not use it.	any

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
J_{mech}	Transmission inertia referred to the motor shaft (screw, pulleys, gearbox, coupling)	0	kg·m²	assumption: Sizing convention.	0 … 10 kg·m²
θ	Incline from horizontal (+ moving uphill)	0	°	assumption: Linear axes only; the rotary table does not use it.	-60 … 90 °
F_p	Process force opposing the move (linear)	0	N	assumption: Linear axes only; the rotary table does not use it.	0 … 1000000 N
T_L	Constant load torque at the output (rotary; continuous duty needs it, and a zero value is refused)	0	N·m	assumption: Sizing convention.	0 … 100000 N·m
J_L	Load inertia at the output (rotary mechanisms)	0.5	kg·m²	spec: S1-style inertia calc of the 600 mm table with fixtures: 0.5 kg·m²	0 … 10000 kg·m²
i	Gear ratio (motor : output, 1 = none)	10		spec: Gearbox datasheet PG-10: 10:1, 85 %	1 … 1000
η	Mechanism efficiency (overall)	0.85		spec: Gearbox datasheet PG-10: 10:1, 85 %	0.3 … 1
μ	Friction coefficient (guide / bearing)	0.1		assumption: Linear axes only; the rotary table does not use it, so the template default is kept.	0 … 1
SF	Safety factor on torque	1.25		assumption: Sizing convention.	1 … 3
$J_{ratio,max}$	Inertia ratio limit (load : motor), warning	10		assumption: Sizing convention.	1 … 100
s	Move distance (linear: m; rotary: output revolutions)	0.25		spec: Cycle specification CS-7: 90° index in 0.3 s, 0.7 s dwell	0.0001 … 10000
t_{move}	Move time	0.3	s	spec: Cycle specification CS-7: 90° index in 0.3 s, 0.7 s dwell	0.01 … 3600 s

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
f_{acc}	Acceleration (and deceleration) fraction of the move time	0.333333		spec: Cycle specification CS-7: 90° index in 0.3 s, 0.7 s dwell	0.05 … 0.5
t_{dwell}	Dwell between moves	0.7	s	spec: Cycle specification CS-7: 90° index in 0.3 s, 0.7 s dwell	0 … 3600 s
J_{motor}	Candidate motor rotor inertia	0.0005	kg·m²	spec: Motor datasheet SM-060: 5.0e-4 kg·m², 3.0 / 9.0 N·m, 3000 rpm	1e-7 … 10 kg·m²
T_{rated}	Candidate motor continuous (rated) torque	3	N·m	spec: Motor datasheet SM-060: 5.0e-4 kg·m², 3.0 / 9.0 N·m, 3000 rpm	0.01 … 100000 N·m
T_{peak}	Candidate motor peak torque	9	N·m	spec: Motor datasheet SM-060: 5.0e-4 kg·m², 3.0 / 9.0 N·m, 3000 rpm	0.01 … 100000 N·m
n_{max}	Candidate motor maximum speed	3000	rpm	spec: Motor datasheet SM-060: 5.0e-4 kg·m², 3.0 / 9.0 N·m, 3000 rpm	10 … 100000 rpm

3. Assumptions

A1. The transmission is rigid. (Servo-sizing practice.)
A2. The mechanism efficiency applies to the driving direction and is also used in the deceleration phase. (Servo-sizing practice.)
A3. This is conservative for the peak and slightly pessimistic for the RMS torque. (Servo-sizing practice.)
A4. The friction force is Coulomb friction on the moved mass. (Load model.)
A5. Viscous and breakaway friction are not modelled unless included in F_p. (Load model.)
A6. In continuous duty the load torque is constant at the stated speed. (Run-up idealisation for pumps and fans.)
A7. The motor accelerates the whole inertia linearly to speed in t_{start} while carrying the full load torque, which is conservative. (Run-up idealisation for pumps and fans.)
A8. Continuous duty ignores the move distance, move time, acceleration fraction, dwell and profile. (Run-up idealisation for pumps and fans.)
A9. The S-curve profile has sine-shaped acceleration over the same acceleration time. (Jerk-limited profile idealisation.)
A10. The peak acceleration of the S-curve profile is π/2 times the trapezoidal value. (Jerk-limited profile idealisation.)
A11. mechanism (Mechanism) is taken as gearbox rotary. (Sizing convention.)
A12. profile (Motion profile) is taken as trapezoidal. (Sizing convention.)
A13. T_{basis} (Design torque basis for downstream couplings (T_{design}): the load's peak (the larger of the accelerating and the braking torque, in magnitude), the motor's typed peak torque, a service factor on the rated torque, or a protective limit such as a torque limiter) is taken as Load peak (T_{pk}). (Design torque for the coupling taken at the load peak; template default.)

A14. brake (A brake or a self-locking drive holds a linear axis during the dwell (no holding torque); unticked, a vertical or inclined axis is held by the motor) is taken as no. (Linear axes only; the rotary table does not use it.)
A15. J_{mech} (Transmission inertia referred to the motor shaft (screw, pulleys, gearbox, coupling)) is taken as 0 kg·m². (Sizing convention.)
A16. θ (Incline from horizontal (+ moving uphill)) is taken as 0 °. (Linear axes only; the rotary table does not use it.)
A17. F_p (Process force opposing the move (linear)) is taken as 0 N. (Linear axes only; the rotary table does not use it.)
A18. T_L (Constant load torque at the output (rotary; continuous duty needs it, and a zero value is refused)) is taken as 0 N·m. (Sizing convention.)
A19. μ (Friction coefficient (guide / bearing)) is taken as 0.1. (Linear axes only; the rotary table does not use it, so the template default is kept.)
A20. SF (Safety factor on torque) is taken as 1.25. (Sizing convention.)
A21. $J_{ratio,max}$ (Inertia ratio limit (load : motor), warning) is taken as 10. (Sizing convention.)

4. Working

[21] $k = 1 / i$
$= 1 / 10$
$= 0.1$
Output radians per motor radian

[22] $T_{hold} = 0\ N*m$
$= 0\ N·m$
Holding torque in the dwell: none for a rotary mechanism (a constant load torque that holds against gravity is not distinguished from friction)

[23] $T_{load} = T_L / (i * \eta)$
$= 0\ N·m / (10 \times 0.85)$
$= 0\ N·m$
Load torque at the motor shaft

[24] $J_{refl} = J_L / (i^2 * \eta)$
$= 0.5\ kg·m^2 / (10^2 \times 0.85)$
$= 0.0058824\ kg·m^2$
Load inertia reflected to the motor shaft

[25] $\theta_m = s * 2 * pi * i$
$= 0.25 \times 2 \times \pi \times 10$
$= 15.708\ rad$
Motor angle for the move

[26] $\omega_{pk} = \theta_m / (t_{move} * (1 - f_{acc}))$
$= 15.708\ rad / (0.3\ s \times (1 - 0.33333))$
$= 78.54\ rad/s$
Peak motor speed (trapezoid)

[27] $\alpha_{trap} = \omega_{pk} / (f_{acc} * t_{move})$
$= 78.54\ rad/s / (0.33333 \times 0.3\ s)$
$= 785.4\ rad/s^2$
Trapezoidal angular acceleration at the motor

[29] $\alpha_{pk} = \alpha_{trap}$
$= 785.4\ rad/s^2$
$= 785.4\ rad/s^2$
Peak acceleration

[30] $\alpha_{pk} = \alpha_{pk}$
$= 785.4\ rad/s^2$
$= 785.4\ rad/s^2$
Peak motor angular acceleration

[31] $J_{total} = J_{motor} + J_{mech} + J_{refl}$
$= 0.0005\ kg·m^2 + 0\ kg·m^2 + 0.0058824\ kg·m^2$
$= 0.0063824\ kg·m^2$
Total inertia at the motor shaft
[32] $T_{acc} = J_{total} * \alpha_{pk}$
$= 0.0063824\ kg·m^2 \times 785.4\ rad/s^2$
$= 5.0127\ N·m$
Acceleration torque
[33] $T_{pk} = T_{acc} + T_{load}$
$= 5.0127\ N·m + 0\ N·m$
$= 5.0127\ N·m$
Peak torque (accelerating against the load)
[34] $T_{dec} = -T_{acc} + T_{load}$
$= -5.0127\ N·m + 0\ N·m$
$= -5.0127\ N·m$
Torque during deceleration
[35] $T_{alt} = (T_{pk} - T_{dec}) / 2$
$= (5.0127\ N·m - (-5.0127\ N·m)) / 2$
$= 5.0127\ N·m$
Alternating torque amplitude of the cycle about T_{load}
[37] $T_{design} = \max(\text{abs}(T_{pk}), \text{abs}(T_{dec}))$
$= \max(\text{abs}(5.0127\ N·m), \text{abs}(-5.0127\ N·m))$
$= 5.0127\ N·m$
Design torque: the load's peak torque, the larger in magnitude of the accelerating T_{pk} and the braking T_{dec} (an overhauling drive brakes harder than it accelerates)
[38] $F_{ax,pk} = 0\ N$
$= 0\ N$
Peak axial thrust on the bearings (none for this mechanism)
[39] $n_{pk} = \omega_{pk}$
$= 78.54\ rad/s$
$= 750\ rpm$
Peak motor speed
[40] $T_{out,m} = T_{load} * i * \eta$
$= 0\ N·m \times 10 \times 0.85$
$= 0\ N·m$
Mean torque on the output shaft (the constant load torque)
[41] $T_{out,a} = T_{alt} * i * \eta$
$= 5.0127\ N·m \times 10 \times 0.85$
$= 42.608\ N·m$
Alternating torque amplitude on the output shaft
[42] $T_{out,pk} = T_{pk} * i * \eta$
$= 5.0127\ N·m \times 10 \times 0.85$
$= 42.608\ N·m$
Peak torque on the output shaft (the motor rotor's inertia torque is counted, so conservative)
[43] $n_{out} = n_{pk} / i$
$= 750\ rpm / 10$
$= 75\ rpm$
Output shaft speed
[44] $P_{pk} = T_{pk} * \omega_{pk}$
$= 5.0127\ N·m \times 78.54\ rad/s$
$= 0.3937\ kW$
Peak mechanical power
[45] $P_{brake} = \max(0\ N*m, -T_{dec}) * \omega_{pk}$
$= \max(0\ N*m, --5.0127\ N·m) \times 78.54\ rad/s$
$= 0.3937\ kW$

Peak braking power at the start of the deceleration

[46] $t_{acc} = f_{acc} * t_{move}$

$= 0.33333 \times 0.3\ s$

$= 0.1\ s$

Acceleration time

[47] $t_{const} = t_{move} - 2 * t_{acc}$

$= 0.3\ s - 2 \times 0.1\ s$

$= 0.1\ s$

Constant-velocity time

[48] $t_{cycle} = t_{move} + t_{dwell}$

$= 0.3\ s + 0.7\ s$

$= 1\ s$

Cycle time

[49] $n_{mean} = \omega_{pk} * (t_{move} - t_{acc}) / t_{cycle}$

$= 78.54\ rad/s \times (0.3\ s - 0.1\ s) / 1\ s$

$= 150\ rpm$

Mean motor speed over the cycle, dwell included (the travel's motor turns per cycle time)

[50] $P_{rms} = \omega_{pk} * sqrt((t_{acc} * (2 / 3) * (T_{load}^2 + T_{acc}^2) + T_{load}^2 * t_{const}) / t_{cycle})$

$= 78.54\ rad/s \times sqrt((0.1\ s \times 2 / 3 \times ((0\ N·m)^2 + (5.0127\ N·m)^2) + (0\ N·m)^2 \times 0.1\ s) / 1\ s)$

$= 0.10165\ kW$

RMS mechanical power over the cycle (thermal-equivalent continuous power)

[51] $T_{rms} = sqrt((T_{pk}^2 * t_{acc} + T_{load}^2 * t_{const} + T_{dec}^2 * t_{acc} + T_{hold}^2 * t_{dwell}) / t_{cycle})$

$= sqrt(((5.0127\ N·m)^2 \times 0.1\ s + (0\ N·m)^2 \times 0.1\ s + (-5.0127\ N·m)^2 \times 0.1\ s + (0\ N·m)^2 \times 0.7\ s) / 1\ s)$

$= 2.2417\ N·m$

RMS torque over the cycle (thermal), the dwell at the holding torque

[52] $J_{ratio} = (J_{refl} + J_{mech}) / J_{motor}$

$= (0.0058824\ kg·m^2 + 0\ kg·m^2) / 0.0005\ kg·m^2$

$= 11.765$

Inertia ratio (load : motor)

[53] $T_{pk,sf} = SF * max(abs(T_{pk}), abs(T_{dec}))$

$= 1.25 \times max(abs(5.0127\ N·m), abs(-5.0127\ N·m))$

$= 6.2659\ N·m$

Peak torque with the safety factor, on the larger of the accelerating and the braking torque

[54] $T_{rms,sf} = SF * T_{rms}$

$= 1.25 \times 2.2417\ N·m$

$= 2.8022\ N·m$

RMS torque with the safety factor

[55] $u_{peak} = SF * max(abs(T_{pk}), abs(T_{dec})) / T_{peak}$

$= 1.25 \times max(abs(5.0127\ N·m), abs(-5.0127\ N·m)) / 9\ N·m$

$= 0.69621$

Peak torque utilisation with safety factor, on the larger of the accelerating and the braking torque

[56] $u_{rms} = SF * T_{rms} / T_{rated}$

$= 1.25 \times 2.2417\ N·m / 3\ N·m$

$= 0.93406$

Continuous torque utilisation with safety factor

[57] $u_{speed} = n_{pk} / n_{max}$

$= 750\ rpm / 3000\ rpm$

$= 0.25$

Speed utilisation

[58] duty = T_{rms} / T_{rated}
= 2.2417 N·m / 3 N·m
= 0.74725
Thermal duty (without margin)

5. Results

SYMBOL	VALUE	DESCRIPTION
T_{load}	0 N·m	Load torque at motor
J_{refl}	0.005882 kg·m²	Reflected load inertia
J_{total}	0.006382 kg·m²	Total inertia
α_{pk}	785.40 rad/s²	Motor angular acceleration
T_{acc}	5.013 N·m	Acceleration torque
T_{pk}	5.013 N·m	Peak torque
T_{dec}	-5.013 N·m	Deceleration torque
T_{rms}	2.242 N·m	RMS torque
T_{design}	5.013 N·m	Design torque for downstream couplings
T_{hold}	0 N·m	Holding torque in the dwell
T_{alt}	5.013 N·m	Alternating torque amplitude
$F_{ax,pk}$	0 N	Peak axial thrust on a screw's bearings
$T_{out,m}$	0 N·m	Output shaft mean torque
$T_{out,a}$	42.61 N·m	Output shaft alternating torque
$T_{out,pk}$	42.61 N·m	Output shaft peak torque
n_{out}	75 rpm	Output shaft speed
n_{pk}	750 rpm	Peak speed
n_{mean}	150 rpm	Mean motor speed over the cycle
P_{pk}	0.3937 kW	Peak power
P_{brake}	0.3937 kW	Peak braking power
P_{rms}	0.1017 kW	RMS power (continuous equivalent)
J_{ratio}	11.765	Inertia ratio
duty	0.74725	Thermal duty
u_{peak}	0.70	Peak utilisation
u_{rms}	0.93	Continuous utilisation (governing)
u_{speed}	0.25	Speed utilisation
k	0.1	Transmission constant
θ_m	15.708 rad	Motor angle
a_{pk}	785.40 rad/s²	Peak acceleration

6. Checks (governing first)

RESULT	CHECK	DEMAND	CAPACITY	UTILISATION	REF
PASS	RMS torque with the safety factor against the rated torque	2.802 N·m	3 N·m	0.93	
PASS	Peak torque with the safety factor against the motor peak torque	6.266 N·m	9 N·m	0.70	
PASS	Peak speed against the maximum speed	750 rpm	3000 rpm	0.25	

7. Applicability

WARNING records: tabulated values not taken from a reference record are J_{motor}, T_{rated}, T_{peak}, n_{max} (Candidate motor rating). Enter each as one so its source, edition and licence travel with it. [record]
WARNING J_{motor}: the inertia ratio 11.8 exceeds 10. [inertia-ratio]
WARNING T_{dec}: the motor brakes during deceleration (negative torque). The drive must absorb the regenerated energy. [overhauling]

8. Verification

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Template elec.motor-sizing@1.11.1 Content hash 7ccfbf19d3cec269 Engine calc-engine/1.0.0
Known-answer cases passed 3/3 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0
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9. References

1. Hughes A, Drury B. Electric Motors and Drives: Fundamentals, Types and Applications. 5th ed. Newnes; 2019.